

Appendix

I. LEARNING PARADIGMS FOR VIDEO AGENTS

The preceding sections describe why agentic control is needed and what video states agents operate on. We now examine how agent behavior is specified or learned. We group existing methods by their dominant mechanism: Training-Free and Inference-Time Control; Supervised Fine-Tuning (SFT) and Imitation Learning, and Reinforcement Learning (RL). These categories are not mutually exclusive, since multi-stage systems may combine supervised initialization with reinforcement learning.

A. Training-Free and Inference-Time Control

Training-free video agents do not optimize a dedicated agent policy. Instead, their behavior is assembled at inference time through prompts, tool interfaces, retrieval procedures, memory rules, verification mechanisms, and stopping or waiting criteria.

The first group follows prompted tool-use and search. VideoAgent [1], OmAgent [2], A4VL [3], VCA [4], LensWalk [5], VideoSeek [6], Deep Video Discovery [7], and Agentic Video Intelligence [8] ask the model to decompose the query, inspect temporal regions or memory entries, call video tools, update intermediate evidence, and synthesize an answer. Their agentic behavior is not learned from trajectories or rewards, but specified through prompts, retrieval modules, external tools, and video states.

Role decomposition can also be implemented in a training-free manner. Symphony [9] separates planning, grounding, subtitle reasoning, visual perception, and reflection into specialized agents. SVAgent [10] assigns visual, textual, and meta-level decision roles for storyline-guided reasoning, while SciEducator [11] organizes scientific video understanding as a Deming-cycle collaboration among agents. These systems show that multi-agent structure can improve modularity and verification even when the roles themselves are prompt-defined rather than learned.

The second group emphasizes inference-time correction and reliability control. DrVideo [12] allows the agent to judge whether the current video document contains sufficient evidence and retrieve missing visual details when needed. VideoAgent2 [13] uses uncertainty-aware reasoning to adjust tool calls. VideoHV-Agent [14] turns candidate answers into hypotheses to be checked against localized evidence. VideoLucy [15] performs confidence-driven memory backtracking, while SEASON [16] uses self-diagnostic contrastive decoding to reduce temporal hallucination. AViLA [17] further extends this paradigm to streaming interaction, where the agent decides whether to answer, retrieve past evidence, or wait for future evidence.

The advantage of this paradigm is accessibility: it can exploit strong foundation models without collecting process level labels. Its weakness is the fragility of the policy. Inference-time agents may over search, stop too early, repeat tool calls, trust noisy tools, or follow fluent but weakly grounded plans.

B. Supervised Fine-Tuning (SFT) and Imitation Learning

Supervised learning trains video-agent behavior at different levels of granularity. Answer-level SFT learns to generate a response from a video and an instruction, but provides limited guidance on how the agent should obtain that response. Imitation learning instead supervises demonstrated decisions or trajectories. In current video agents, it is commonly implemented as SFT on teacher-generated traces; the distinction therefore lies mainly in the supervision target rather than the optimization algorithm.

Trajectory imitation is often used to initialize an agent before preference or reinforcement learning. VideoExplorer [18] explicitly performs structured imitation by training its planner and temporal grounder on successful reasoning trajectories. LongVideo-R1 [19] learns from chain-of-thought-with-tool demonstrations that specify when to inspect a video segment, continue reasoning, or terminate. SAGE [20] similarly constructs tool-use trajectories with a stronger model and converts them into state-action pairs for cold-start SFT. In these systems, imitation establishes a valid initial policy, while later optimization improves its effectiveness.

Other methods apply supervised learning to selected components rather than complete trajectories. VideoMind [21] separately trains planner, grounder, and verifier roles using planning labels, temporal localization objectives, and verification labels, with Chain-of-LoRA enabling role switching at inference. AoTD [22] transfers verified reasoning traces produced by an external agent system into a single Video-LLM, while VITED [23] trains models to generate automatically constructed temporal evidence chains.

Supervised learning offers stable and dense behavioral guidance, but remains constrained by the coverage and quality of its demonstrations or labels. It can teach an agent to reproduce successful patterns without ensuring that it can adapt when unfamiliar states require a different strategy.

C. Reinforcement Learning

Reinforcement learning (RL) optimizes video agent decisions through outcome or process level rewards rather than only imitating demonstrated trajectories. Rewards may measure answer correctness, evidence grounding, action efficiency, response timing, or reasoning validity. This is particularly useful for learning decisions such as where to inspect, which

evidence to retain, when to invoke a tool, and when sufficient evidence has been collected to answer.

One line of work applies RL to perception and evidence acquisition. SAGE [20] trains agents to choose among direct answering, rapid skimming, and deeper video inspection. Thinking with Videos [24] learns tool-augmented policies for acquiring evidence from long videos, while APPO [25] optimizes attention-guided selection of informative visual tokens. LongVideo-R1 [19] learns navigation over hierarchical video structures. R3-Streaming [26] extends such control to streaming settings by optimizing memory forgetting, answer readiness, and fast-slow processing routes.

RL can also optimize coordination among multiple agents. ReAgent-V [27] supports self-correction through a critic agent that provides feedback and generates follow-up questions for the target agent. VideoChat-M1 [28] uses collaborative planning and multi-agent reinforcement learning to optimize how specialized agents propose, execute, and revise tool-use plans. Unlike systems whose coordination is defined primarily through prompts, these methods make coordination decisions part of the learned behavior.

Another line focuses on verifiable rewards for video reasoning and grounding without learning a broader perception-and-tool control policy. Time-R1 [29] uses localization-oriented rewards for temporal grounding, while Video-R1 [30] improves temporal reasoning through RL post-training. DeepVideo-R1 [31] introduces difficulty-aware regressive GRPO, and When Thinking Drifts [32] rewards reasoning trajectories that remain grounded in visual evidence.

RL is well suited to video agents because many decisions lack direct supervision. Its main challenge is credit assignment: sparse rewards often cannot reveal whether success comes from a reliable policy or accidental shortcuts.

II. DATA AND SUPERVISION FOR VIDEO AGENTS

Learning video agents requires supervision that goes beyond video-question-answer pairs. A useful training example may need to show which evidence was inspected, which tool was called, what memory was written, which temporal interval grounded the claim, and why the agent stopped, revised, or responded. This section separates three data-related aspects: trajectory supervision, grounding supervision, and preference and reward supervision, and agent-generated data as a supervision source.

A. Trajectory Supervision

Trajectory supervision records the step-by-step path by which an agent reaches an answer or action. A trace should record the agent’s observations, decisions, state changes, and final response, with richer traces also capturing failed branches, revisions, stopping decisions, and cost. Such data is useful not only for imitation learning, but also for auditing whether a video agent followed a grounded decision path rather than merely producing a plausible final answer.

For video understanding, trajectory supervision is more complex than in text-only agents. The trace should identify the

video units involved in each step, such as frames, clips, transcript spans, audio segments, object tracks, temporal segments, or memory entries, and record how these observations changed the next action. Agent-of-Thoughts Distillation [22] distills tool-supported decomposition and reasoning traces into a video model. ReWatch-R1 [33] bootstraps timestamped, multi-agent Reasoning and Acting (ReAct)-style traces for complex video reasoning. VITED [23] distills temporal evidence chains, while VTimeCoT [34] represents reasoning as an annotated visual temporal canvas. VideoExplorer [18] further organizes long-video reasoning as trajectories that connect planning, temporal grounding, and evidence reading.

Interactive video data adds another layer: the timestamp of each observation, query, interruption, and response may matter. StreamBridge [35] constructs interleaved video-text instruction traces for streaming assistants, Takusen [36] supervises asynchronous dense captioning through event-start and event-end decisions, and ViSpeak [37] includes visual instructions such as wake-up, reference, interruption, and termination.

B. Grounding Supervision

Grounding supervision links claims, answers, intermediate states, and actions to their supporting video evidence. It includes temporal intervals, spatial regions, entity identities, object tracks, audio cues, transcript spans, OCR evidence, and state changes. Temporal evidence alignment is especially important because many video-agent failures arise when an answer is semantically plausible but tied to the wrong moment, or when a reasoning step uses evidence that was never actually observed.

Grounded reasoning systems point toward this requirement. VideoMind [21] grounds reasoning in temporal candidate moments before answer generation. VideoHV Agent [14] turns candidate answers into hypotheses and verifies them against localized video evidence. Refer-Agent [38] extends grounding to spatial evidence by checking object existence, identity consistency, and mask reliability. A4VL [3] aligns query-specific perception clues with relevant video blocks, while MAGNET [39] stresses audio-visual grounding across multi-video haystacks. In the long run, grounding supervision should include negative evidence as well as positive evidence: the agent should know not only which segment supports an answer, but also which inspected segments failed to support competing hypotheses.

Interactive and causal settings further expand grounding from *where is the evidence?* to *when was the evidence available?* AViLA [17] highlights query-evidence asynchrony, StreamEQA [40] attaches questions to timestamps, and Streaming-Eval [41] evaluates punctual behavior. In such settings, a correct answer may still be invalid if it relies on future evidence unavailable at decision time.

C. Preference and Reward Supervision

Preference and reward supervision provides comparative or scalar signals for learning which agent behavior should be preferred. Unlike trajectory supervision, which records the

process, or grounding supervision, which anchors claims to evidence, this form of data evaluates the quality of answers, evidence choices, reasoning traces, tool calls, or complete rollouts.

Recent work illustrates this supervision at different granularities. V-Agent [42] fine-tunes a retrieval-oriented VLM with a small video preference dataset, providing preference signals for video search rather than full agent rollouts. DeepVideo-R1 [31] organizes reward supervision around sample difficulty through regressive GRPO, while Video-R1 [30] introduces temporal-aware rewards to encourage reasoning grounded in video dynamics. ThinkStream [43] derives automatically verifiable rewards for incremental reasoning and watch-think-speak timing. R3-Streaming [26] further uses routing rewards that reflect response readiness, latency, memory pressure, and fast-slow model selection.

The main challenge is reward validity. If rewards only reflect answer correctness, they may miss whether the agent searched efficiently, used reliable evidence, or avoided spurious reasoning.

TABLE S1
TAXONOMIC OVERVIEW OF VIDEO UNDERSTANDING AGENT METHODS (CONTINUED). ADDITIONAL SURVEYED METHODS, ORDERED BY PUBLICATION YEAR.

Method	Year	Challenge	Paradigm	Learning Paradigms			Data and Supervision		
				Training-Free Control	Supervised Fine-Tuning	Reinforcement Learning	Trajectory Supervision	Grounding Supervision	Reward Signals
VCA [4]	2025	Evidence Sparsity	Paradigm II	★	★	★	★	★	★
Video-R1 [30]	2025	Temporal Causality	Paradigm II	★	★	★	★	★	★
Video-RAG [44]	2025	Context Bottleneck	Paradigm III	★	★	★	★	★	★
VideoAgent2 [13]	2025	Evidence Sparsity	Paradigm II	★	★	★	★	★	★
VideoAgentTrek [45]	2025	Temporal Causality	Paradigm II	★	★	★	★	★	★
VideoExplorer [18]	2025	Evidence Sparsity	Paradigm III	★	★	★	★	★	★
VideoLLaMB [46]	2025	Context Bottleneck	Paradigm III	★	★	★	★	★	★
VideoLucy [15]	2025	Context Bottleneck	Paradigm III	★	★	★	★	★	★
ViSpeak [37]	2025	Multimodal Ambiguity	Paradigm IV	★	★	★	★	★	★
VITED [23]	2025	Temporal Causality	Paradigm II	★	★	★	★	★	★
VTimeCoT [34]	2025	Temporal Causality	Paradigm III	★	★	★	★	★	★
When Thinking Drifts [32]	2025	Temporal Causality	Paradigm II	★	★	★	★	★	★
A4VL [3]	2026	Evidence Sparsity	Paradigm I	★	★	★	★	★	★
AgenticVS [47]	2026	Multimodal Ambiguity	Paradigm II	★	★	★	★	★	★
APPO [25]	2026	Evidence Sparsity	Paradigm I	★	★	★	★	★	★
EGAgent [48]	2026	Temporal Causality	Paradigm III	★	★	★	★	★	★
EVA [49]	2026	Evidence Sparsity	Paradigm I	★	★	★	★	★	★
G2F-RAG [50]	2026	Context Bottleneck	Paradigm III	★	★	★	★	★	★
HAVEN [51]	2026	Context Bottleneck	Paradigm III	★	★	★	★	★	★
LensWalk [5]	2026	Evidence Sparsity	Paradigm II	★	★	★	★	★	★
LongVideo-R1 [19]	2026	Evidence Sparsity	Paradigm II	★	★	★	★	★	★
OmniAgent [52]	2026	Evidence Sparsity	Paradigm IV	★	★	★	★	★	★
Proact-VL [53]	2026	Temporal Causality	Paradigm IV	★	★	★	★	★	★
R3-Streaming [26]	2026	Context Bottleneck	Paradigm IV	★	★	★	★	★	★
ThinkStream [43]	2026	Temporal Causality	Paradigm IV	★	★	★	★	★	★
Refer-Agent [38]	2026	Temporal Causality	Paradigm I	★	★	★	★	★	★
ReViSe [54]	2026	Evidence Sparsity	Paradigm I	★	★	★	★	★	★
SAGE [20]	2026	Evidence Sparsity	Paradigm II	★	★	★	★	★	★
StreamAgent [55]	2026	Temporal Causality	Paradigm IV	★	★	★	★	★	★
Streaming Harness [41]	2026	Temporal Causality	Paradigm IV	★	★	★	★	★	★
StreamMeCo [56]	2026	Context Bottleneck	Paradigm III	★	★	★	★	★	★
StreamRAG [57]	2026	Context Bottleneck	Paradigm III	★	★	★	★	★	★
StreamReady [58]	2026	Temporal Causality	Paradigm IV	★	★	★	★	★	★
SVAgent [10]	2026	Temporal Causality	Paradigm III	★	★	★	★	★	★
Symphony [9]	2026	Multimodal Ambiguity	Paradigm II	★	★	★	★	★	★
Takusen [36]	2026	Temporal Causality	Paradigm IV	★	★	★	★	★	★
V-Agent [42]	2026	Multimodal Ambiguity	Paradigm I	★	★	★	★	★	★
Video-MTR [59]	2026	Evidence Sparsity	Paradigm II	★	★	★	★	★	★
VideoARM [60]	2026	Context Bottleneck	Paradigm III	★	★	★	★	★	★
VideoBrain [61]	2026	Evidence Sparsity	Paradigm I	★	★	★	★	★	★
VideoChat-A1 [62]	2026	Evidence Sparsity	Paradigm II	★	★	★	★	★	★
VideoChat-M1 [28]	2026	Multimodal Ambiguity	Paradigm II	★	★	★	★	★	★
VideoHV-Agent [14]	2026	Temporal Causality	Paradigm II	★	★	★	★	★	★
VideoMind [21]	2026	Temporal Causality	Paradigm III	★	★	★	★	★	★
VideoSEAL [63]	2026	Evidence Sparsity	Paradigm II	★	★	★	★	★	★
VideoSeek [6]	2026	Evidence Sparsity	Paradigm II	★	★	★	★	★	★
WorldMM [64]	2026	Context Bottleneck	Paradigm III	★	★	★	★	★	★

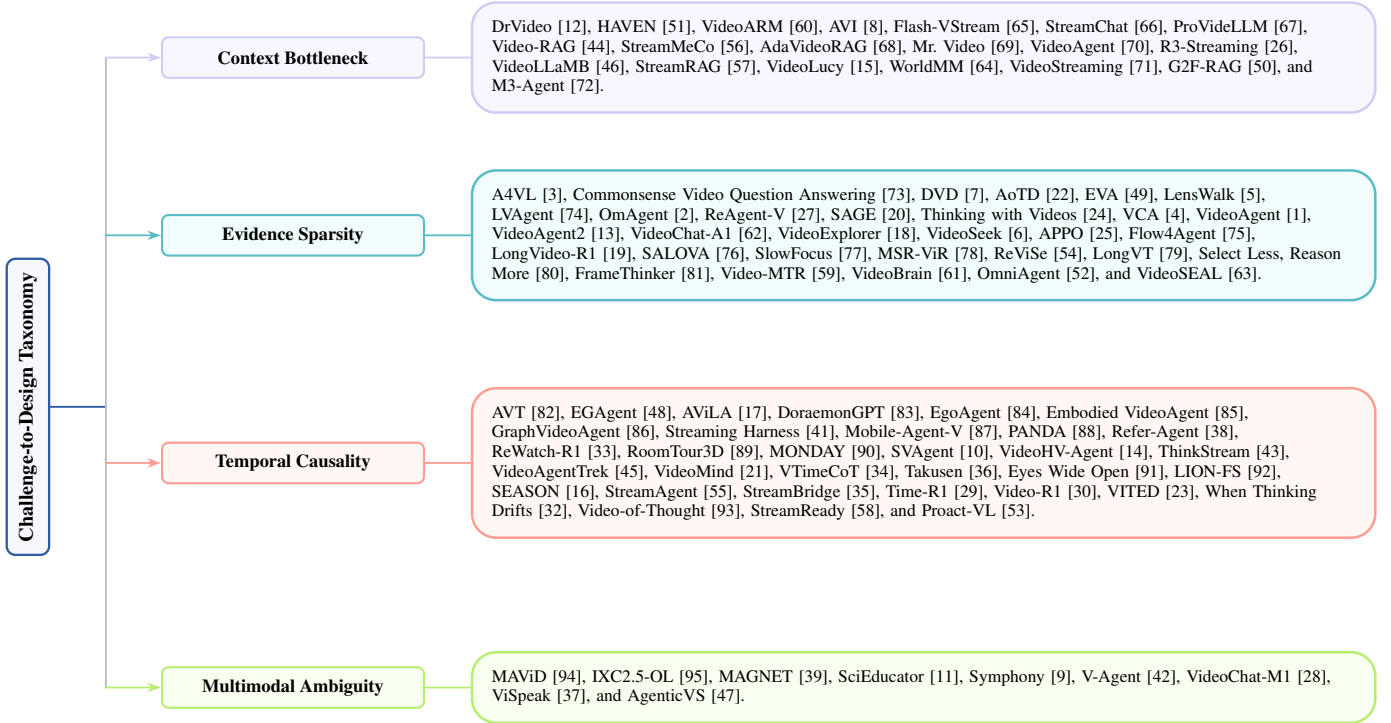


Fig. S1. Challenge-to-design taxonomy of video understanding agents. Each branch links a video-specific bottleneck to representative agentic systems that address it.

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